

Bijay Marasini

MSc DATA SCIENCE | AI

MSc Data Science student specializing in Generative AI, Natural Language Processing, and Retrieval-Augmented Generation (RAG). Research-focused experience in designing, implementing, and evaluating domain-adaptive LLM systems on large-scale datasets. Strong interest in developing robust, explainable, and efficient AI systems for real-world applications.

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SKILLS

Agentic AI & LLM Systems:	LangChain, LangGraph, Multi-Agent Systems, State Machines, LLM APIs (OpenAI)
Retrieval-Augmented Generation:	FAISS, Vector Databases, Semantic Search, Cross-Encoder Reranking, Prompt Engineering,
ML & NLP:	PyTorch, Transformers, Hugging Face, Model Evaluation, TensorFlow, Feature Engineering
Programming:	Python, Java, JavaScript, C++, C#(.NET)
Backend & APIs:	Node.js, Spring Boot, Spring MVC, Flask, FastAPIs, RESTful APIs
Data & Databases:	Pandas, Numpy, PostgreSQL, MySQL, SQL Server, MongoDB, Oracle
Data Visualization & BI Tools:	Matplotlib, Seaborn, Power BI, Tableau
Software Engineering Concepts:	System Design, Performance Optimization, Design Patterns, Code Reviews
DevOps & Tools:	Docker, Git, CI/CD, Linux, Jenkins
Agile Methodology:	Scrum, Kanban
Architecture:	Microservices, Client-Server

EXPERIENCES

Research Assistant - Generative AI, *Institut für Arbeitsmarkt - und Berufsforschung* | Nürnberg, Germany

Oct 2024 - Dec 2025

- Led development of a RAG-based job-matching pipeline using FAISS embeddings, cross-encoder reranking, and LoRA fine-tuning - increased model F1 from 0.26 to 0.80 and improved precision by 60%.
- Built an automated weak-supervision pipeline to generate 41,000 labeled samples, reducing manual labeling effort by 50% and enabling downstream model training improvements.
- Implemented end-to-end evaluation (Precision/Recall/F1/AUC) and monitoring dashboards to track model drift and performance for stakeholders.
- Cleaned, transformed, and analyzed large-scale datasets using Python (Pandas, NumPy) and SQL, and created visual reports to communicate insights and model performance to stakeholders.
- Collaborated with domain experts to ensure ethical, explainable AI practices; produced reproducible training notebooks and model cards for audits.
- Tech Stack** - Python, FAISS, LoRA, PyTorch, sentence-transformers, cross-encoder, SQL, Pandas, Matplotlib, Docker, MLflow.

Working Student - Full-Stack Developer, *A. Eberle GmbH & Co. KG* | Nürnberg, Germany

Jan 2023 - Jul 2023

- Contributed to system design decisions for dashboard architecture and ensured scalability of data visualization services for power quality services.
- Improved user role and permission management, strengthening application security and access control.
- Developed and maintained frontend and backend features using React and Node.js, following clean architecture principles.
- Tech Stack** - Node.js, React, JavaScript, TypeScript, PostgreSQL, Visual Studio Code.

Full-Stack Developer, ICC Loyalty | Noida, India

Feb 2022 - Sept 2022

- Designed, developed, and maintained customer-facing web applications, including a Gift Card Management System and Loyalty Points Program, leveraging data analytics to improve customer engagement and retention.
- Integrated secure Visa and Mastercard payment gateways enabling real-time online transactions.
- Implemented MFA authentication, strengthening security compliance and user data protection.
- Optimized API performance through query tuning and efficient data modeling, reducing response latency.
- Tech Stack** - Java, Spring Boot, MySQL, Hibernate, Spring MVC, jQuery, AJAX, JSON, JSP, NetBeans.

Software Engineer - Product Development, *Capgemini* | Gurugram, India

Dec 2019 - Jan 2022

- Developed telecom product workflows including custom data/voice packs and bundle offers to support customer personalization.
- Designed USSD flows for real-time activation and recharge automation.
- Implemented automated notification systems to alert customers on recharge success, package expiration, and 80% data consumption thresholds, improving user experience and transparency.
- Participated in system design discussions, code reviews, and applied design patterns to build scalable and maintainable telecom applications.
- Optimized backend performance using database tuning, caching, and asynchronous processing.
- Tech Stack** - Java, Spring Boot, SpringMVC, PostgreSQL, Maven, Eclipse, JaCoCo, Mockito, JMS API, RabbitMQ, XML, Unit Testing, Integration Testing.

SELECTED PROJECTS

ArguMind - Multi-Agent AI Debate System

- Designed and developed a full-stack Multi-Agent AI system using LangGraph to orchestrate Pro, Contra, Judge, and Critic agents in a state-machine workflow.
- Developed a FastAPI backend exposing REST endpoints to run debates and return structured AI outputs.
- Created an interactive React.js + React Flow frontend to visualize agent execution as draggable, real-time graph nodes.
- Implemented SOLID architecture and dependency injection for modular, extensible, and testable agent design.
- **Tech Stack** -Python, LangGraph, LangChain, Google Gemini API, FastAPI, React.js, React Flow, JavaScript, REST APIs, Docker (optional), Git, JSON, SOLID Principles, Agentic AI, State Machines.

URL: github.com/BIJAYS09/ArguMind

RAG-Based AI Chatbot

- Developed a Python-based Retrieval-Augmented Generation (RAG) chatbot application to deliver context-aware responses by combining information retrieval with large language model generation.
- Integrated third-party LLM APIs and supported libraries via a structured requirements.txt, ensuring reproducible dependency management.
- Leveraged semantic vector search techniques for effective retrieval of relevant response context, reducing hallucinations and improving answer relevance.
- Designed and containerized the application using Docker and Docker Compose for efficient environment setup, deployment, and scalability.
- **Tech Stack** -Python, Docker, Docker Compose, RAG (Retrieval-Augmented Generation), Semantic Search, LLM APIs, Vector Databases / Embeddings, Web Framework, Dependency Management.

URL: github.com/BIJAYS09/RAG-Based-AI-Chatbot

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg, MSc Data Science | Erlangen, Germany Oct 2022 - Mar 2026

Thesis: Domain-adaptive RAG for job matching in labor market datasets.

- Coursework: Deep Learning, Pattern Recognition, Machine Learning for Time Series, Machine Learning in Finance, Business Intelligence, Explainable Machine Learning, Knowledge in Database Discovery, Selected Topics in Machine Learning.

Aug 2015 - Jul 2019

Guru Gobind Singh Indraprastha University, B.Tech in Information Technology | Delhi, India

- Coursework: Software Engineering, Applied Mathematics, Web Technology, Data Structures & Algorithms, Object Oriented Programming, Advanced Database Administration.

LANGUAGES

English	Professional proficiency (C1)
German	Intermediate proficiency (A2, Actively Learning)
Hindi	Native proficiency